UNIVERSITY OF SCIENCE AND TECHNOLOGY
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Lecture Activity 3

IT325 Machine Learning | Comparing Models and Reading Evaluation Metrics

Type	Lecture Activity
Topic Focus	Model comparison and evaluation for classification
Time Allotment	45 to 60 minutes
PIT Connection	Students learn how to justify model choice using evidence
Mode	Individual or pair discussion

Learning Outcomes

1. Explain why more than one model should be tested in a classification problem.
2. Differentiate accuracy, precision, recall, F1 score, and confusion matrix results.
3. Interpret model evaluation results and justify an initial best model.

Activity

Students analyze sample evaluation results from several classifiers and decide which model is most suitable for a given text classification task. The emphasis is on reasoning, not yet on coding.

Suggested Discussion Points

1. Why using only one model may lead to a weak PIT result

Using only one model may lead to a weak result because different models have different strengths and weaknesses. For example, Logistic Regression might perform well on linearly separable patterns, but it may miss more complex relationships in the data. If I only test one model and it happens to have a decent accuracy, I might think it is good enough when in reality a different model like Random Forest could perform much better. Testing multiple models lets me compare fairly and pick the one that actually works best for my dataset.

2. Why accuracy alone is not enough in many classification tasks

Accuracy tells me how many predictions were correct overall, but it can be misleading when the data is imbalanced. In my Student Depression Dataset, about **58.5%** of students are depressed. If a model just predicts everyone as depressed, it would already get **58.5%** accuracy without actually learning anything useful. This is why I also need to look at precision, recall, and F1 score to get a clearer picture of model performance.

3. How false positives and false negatives affect decision making

A false positive means the model predicted a student is depressed when they are actually not. A false negative means the model predicted a student is not depressed when they actually are. In my dataset, false negatives are more dangerous because missing a student who is truly depressed means they do not get help. So I want a model with high recall, meaning it catches as many actual depressed students as possible, even if it sometimes gives a few false alarms.

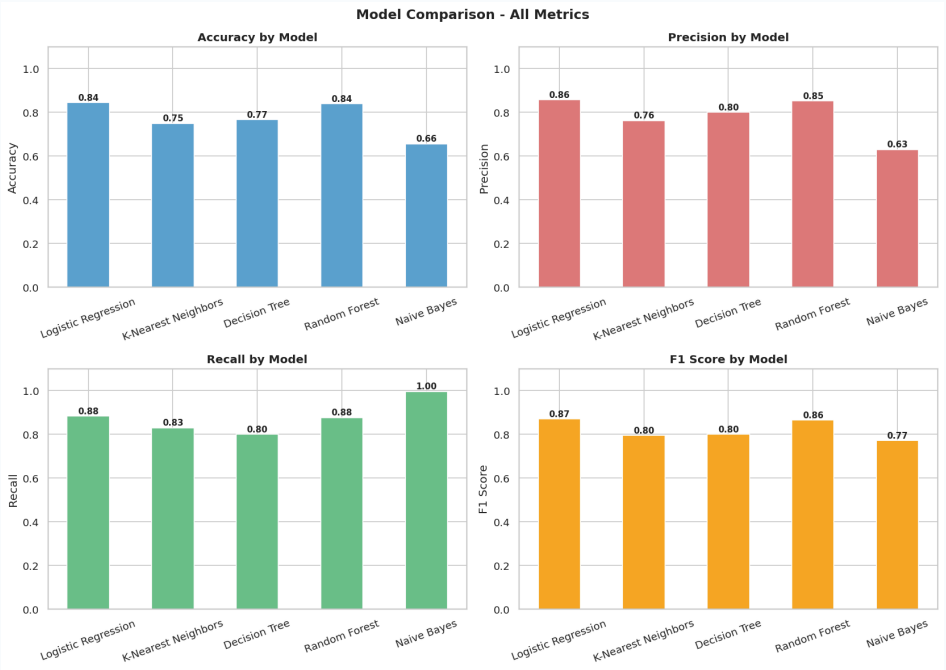
4. How evaluation metrics help in choosing models fairly

Evaluation metrics like precision, recall, F1 score, and the confusion matrix help me compare models based on more than just accuracy. For example, if Model A has 85% accuracy but low recall, and Model B has 83% accuracy but high recall, I would choose Model B for my depression detection task because missing real cases is worse than a slightly lower overall accuracy. These metrics make the comparison fair and evidence-based.

In Class Analysis Task

1. Review a sample table of results from five models.

Model Comparison Table (Student Depression Dataset - Actual Results):



2. Identify the model with the highest accuracy.

Logistic Regression has the highest accuracy at 84.46%.

3. Identify the model with the highest recall.

Naive Bayes has the highest recall at 0.9960. This means it is the best at catching actual depressed students, even though its overall accuracy is only 65.57%, the lowest among all five models.

4. Identify the model with the most balanced performance.

Logistic Regression has the most balanced performance. It has the highest accuracy (84.46%), strong precision (0.8567), high recall (0.8822), and the highest F1 score (0.8693). This means it performs consistently well across all metrics without sacrificing one for another.

5. Write a short justification for the model you would carry forward for tuning.

I would carry forward Logistic Regression and Random Forest for tuning. Logistic Regression achieved the best overall performance with **84.46%** accuracy and an F1 score of **0.8693**, making it the most balanced model.

Random Forest came in second with **83.87%** accuracy and an F1 score of **0.8643**, which is very close and still reliable across all metrics. Both models significantly outperformed K-Nearest Neighbors, Decision Tree, and Naive Bayes in balanced performance. While Naive Bayes had the highest recall at **0.9960**, its accuracy of only **65.57%** makes it too unreliable for practical use. I will tune Logistic Regression and Random Forest to try to improve their recall further.

Reflection Prompt

A model with the highest accuracy is not always the best model. Explain this statement using a classification example related to your PIT dataset.

In my Student Depression Dataset, Naive Bayes had the highest recall at **0.9960**, meaning it almost never missed a depressed student. However, its accuracy was only **65.57%**, the lowest among all five models. This means it was incorrectly flagging too many non-depressed students as depressed. If I only looked at accuracy, I would pick Logistic Regression at **84.48%** and potentially overlook that Naive Bayes catches almost every real case of depression.

In a mental health setting, missing a depressed student (false negative) can be more harmful than a false alarm (false positive). So depending on the goal, the best model is not always the one with the highest accuracy. I need to look at all the metrics together, especially recall and F1 score, before making a final decision.